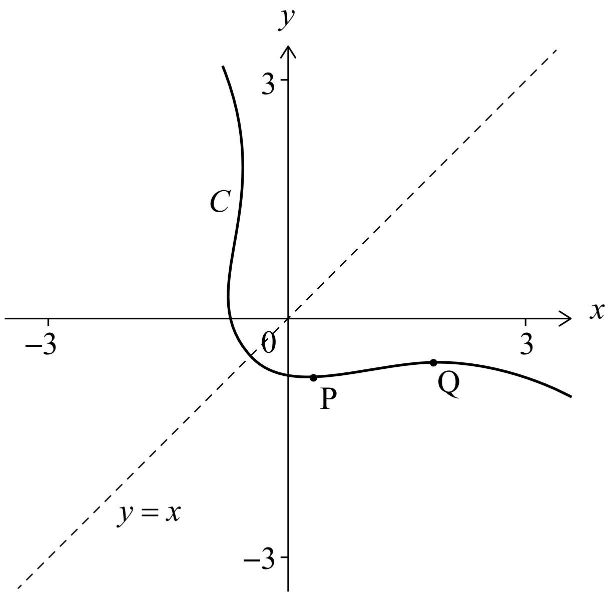
Do **not** write solutions on this page.

11. [Maximum mark: 19]

Consider the curve C defined by the equation $e^{x+y} = x^2 + y^2$, shown on the following diagram. The curve has a line of symmetry $y = x$.

There are two points on the curve C where the tangent is horizontal. These points are labelled P and Q.



(a) Show that $\dfrac{dy}{dx} = \dfrac{2x - e^{x+y}}{e^{x+y} - 2y}$. [5]

(b) (i) Show that the x-coordinates of points P and Q satisfy the equation $2x^2 + (\ln(2x))^2 - 2x\ln(2x) - 2x = 0$.

(ii) Hence, find the coordinates of P and the coordinates of Q. [9]

(c) Using the line of symmetry, write down the coordinates of the points on the curve C where the tangent is vertical. [1]

(d) Find the coordinates of the point on the curve C where the tangent has a gradient of -1. [4]